

Highly Indistinguishable Dual-Channel Integrated Microring Single-Photon Source

Baojie Hou⁽¹⁾, Donghui Chen⁽¹⁾, Haoran Ma⁽¹⁾, Zichao Zhao⁽¹⁾, Qishen Liang⁽¹⁾, Xinyu Wang^(1,2), Bin Zhang^(1,2), Yawen Tu⁽¹⁾, Yongdi Zhang⁽¹⁾, Bangmin Gong⁽¹⁾, Tingge Dai^(1,3,*), Yuehai Wang⁽¹⁾, Huihui Zhu^(1,*), Jianyi Yang^(1,4,*)

⁽¹⁾ College of Information Science and Electronic Engineering and ZJU-Hangzhou Global Scientific and Technological Innovation Center, Zhejiang University, Hangzhou, Zhejiang, China, huihui001@zju.edu.cn

⁽²⁾ The Zhejiang Lab, Hangzhou, 311121, China

⁽³⁾ Ningbo Global Innovation Center, Zhejiang University, Ningbo 315100, China, daitingge@zju.edu.cn

⁽⁴⁾ Jinhua Institute of Zhejiang University, Jinhua, 321000, China, yangjy@zju.edu.cn

Abstract Here, we propose a highly indistinguishable dual-channel integrated microring single-photon source based on bidirectional pumping. It achieves spectral purities of 96.13% and 96.96% in the two directions, respectively, alongside a 95.89% two-photon interference visibility. ©2026 The Author(s)

Introduction

Integrated single-photon sources are fundamental building blocks for scalable quantum information processing, playing a pivotal role in advanced quantum communication networks and multi-photon interference tasks [1]. To execute high-fidelity quantum operations, these sources must generate photons that are not only highly pure but also strictly indistinguishable. Among various integrated platforms, silicon micro-ring resonators (MRRs) have emerged as highly attractive candidates. By leveraging cavity resonance enhancement and spontaneous four-wave mixing (SFWM), MRRs offer high-brightness photon generation within a compact, CMOS-compatible footprint [2].

Despite these advantages, standard MRR single-photon sources face a theoretical spectral purity limit of approximately 92% [3, 4]. Recently, on-chip dual-pulse control has successfully broken this barrier, achieving purities exceeding 98% [5]. However, scaling this high-purity scheme to multi-photon systems faces two critical challenges. First, unavoidable fabrication tolerances induce structural variations between spatially separated MRRs, causing resonance mismatches that severely degrade photon indistinguishability [6]. Correcting these mismatches typically requires active thermal tuning, which introduces unwanted power consumption, system complexity, and thermal crosstalk. Second, the dual-pulse shaping scheme inherently suffers from a reduced photon generation rate, limiting overall system efficiency [7, 8].

To overcome these limitations simultaneously, we propose and experimentally demonstrate a highly indistinguishable, dual-channel integrated MRR single-photon source based on a bidirectional pumping architecture [9]. By exploiting the

intrinsic symmetry of a single MRR pumped concurrently in the clockwise and counter-clockwise directions, our approach inherently guarantees identical resonance frequencies, eliminating the need for complex active tuning. Fabricated on a passive silicon platform, the device achieves bidirectional spectral purities of 96.13% and 96.96%. Furthermore, we demonstrate a two-photon interference visibility of 95.89%. Crucially, by spatially extracting the dual-channel outputs, this architecture effectively mitigates the rate-loss penalty, providing a robust and highly efficient solution for large-scale on-chip quantum networks.

Experimental Setup and Device Design

The proposed integrated quantum photonic device, fabricated on a passive silicon-on-insulator (SOI) platform with standard 450 nm × 220 nm waveguides, is illustrated in Figure 1. The comprehensive device architecture is detailed in the overall schematic (Fig. 1(a)), which is partitioned into three functional modules: (A) Pulse Shaping, (B) Single-photon source Generation and Filtering, and (C) Quantum Interference. As shown in Fig. 1(a), the TE-polarized pump light is coupled into the chip via a grating coupler. To ensure high spectral purity, a dual-pulse pumping scheme is implemented within the pulse shaping module, where a relative time delay of $\Delta\tau = 60$ ps is generated by an on-chip MZI with a delay line [5]. The core of the device is the bidirectional pumping architecture for photon-pair generation, explicitly detailed in Fig. 1. By exploiting the intrinsic symmetry of a single microring resonator (MRR) positioned on the symmetry axis and pumped concurrently in both clockwise (CW) and counter-clockwise (CCW) directions, this approach inherently guarantees identical resonance frequencies without requiring active resonance alignment. Within the MRR, photon

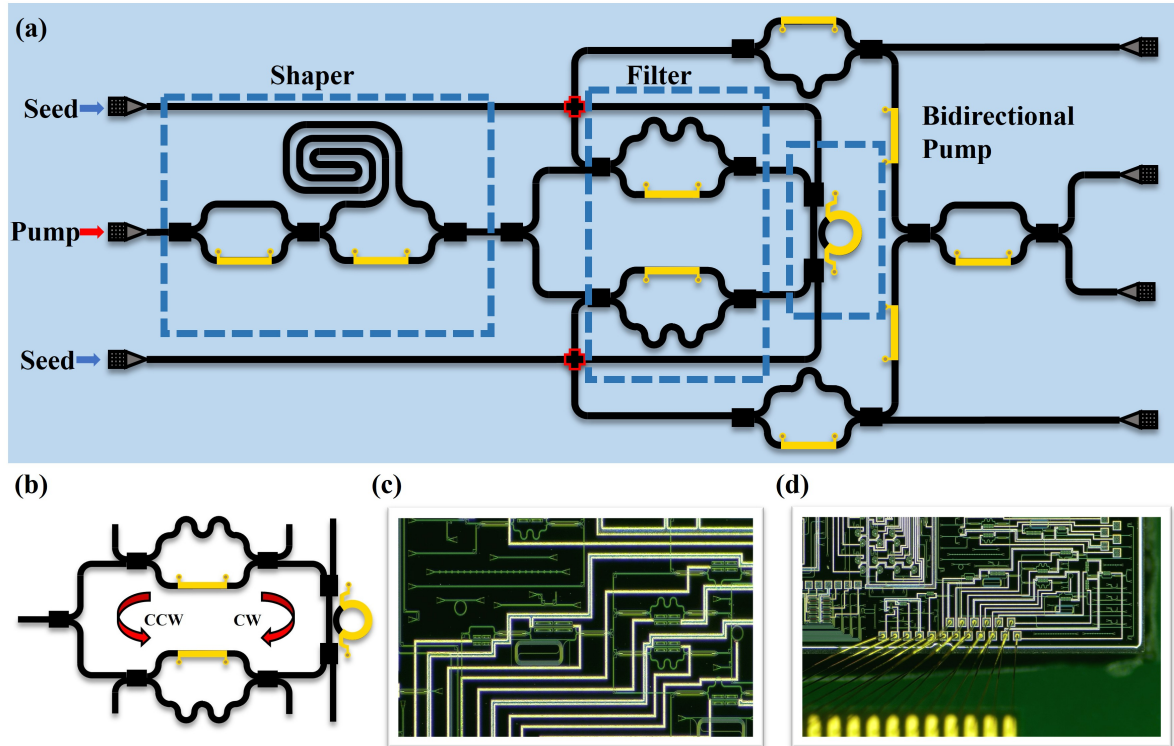


Fig. 1: Device design of the highly indistinguishable integrated single-photon source. (a) Schematic of the integrated quantum photonic chip structure, which comprises the pulse shaper, filter, and bidirectional pump modules. (b) Detailed schematic of the bidirectional pumping structure, illustrating the concurrent clockwise (CW) and counter-clockwise (CCW) excitation within a single microring resonator. (c) Optical microscope image of the fabricated silicon chip. (d) Photograph of the fully packaged device.

pairs are generated via spontaneous four-wave mixing (SFWM) and separated from the residual pump by specific on-chip filtering networks. The fabricated chip is characterized via optical microscopy (Fig. 1(c)), where the real device layout clearly shows the three functional regions: "Pulse Shaping," "Bidirectional Excitation," and "Quantum Interference." Finally, the extracted photons enter a reconfigurable quantum interference network (as shown in Fig. 1(a)) to observe two-photon interference. The output photons are coupled off-chip via grating couplers. A photo of the packaged device (Fig. 1(d)) shows the chip mounted on a PCB, demonstrating the complete packaged device.

Results

The performance of the single-photon source is characterized by tuning the microring resonance to the central point of the shaped pump pulse. Under the bidirectional pumping scheme, the extracted photon pairs from each direction are first passed through cascaded dense wavelength division multiplexers (DWDMs) to deeply reject the residual pump light. Subsequently, the signal photons are routed into a 50:50 fiber beam splitter for coincidence counting (note that including the idler photons for multi-fold coincidence counting enables a heralded Hanbury Brown and Twiss (HBT) experiment). The measured unheralded second-order correlation functions, $g_u^{(2)}(\Delta t)$, for both directions are presented in Fig. 2(a) and Fig. 2(b). The zero-

delay values, $g_u^{(2)}(0)$, reach 1.9613 ± 0.016 and 1.9696 ± 0.016 , respectively. Both values exceed 1.96, corresponding to exceptional spectral purities of 96.13% and 96.96%. These results indicate that the emitted unheralded signal photons exhibit a highly thermal distribution, demonstrating excellent high-purity consistency between the two bidirectional channels.

Furthermore, to characterize the single-photon nature of the generated sources, we performed a heralded HBT experiment. By routing the idler channel to the detector for three-fold coincidence measurements, we recorded the coincidence counts between the split signal channels and the idler channel, the three-fold coincidence counts, and the single counts of the idler channel. The heralded second-order correlation functions, $g_h^{(2)}(\Delta t)$, for both directions are calculated accordingly and plotted in Fig. 2(c) and Fig. 2(d). The measured zero-delay heralded correlation values are $g_h^{(2)}(0) = 0.0650 \pm 0.039$ for the CW direction and $g_h^{(2)}(0) = 0.0488 \pm 0.043$ for the CCW direction. Both values are significantly below the multiphoton threshold of 0.5, corresponding to high photon-number purities of 96.75% and 97.56%, respectively. These results confirm that the photons emitted from both directions can be confidently regarded as high-quality single-photon emissions, with negligible multi-photon events at the applied pump power level.

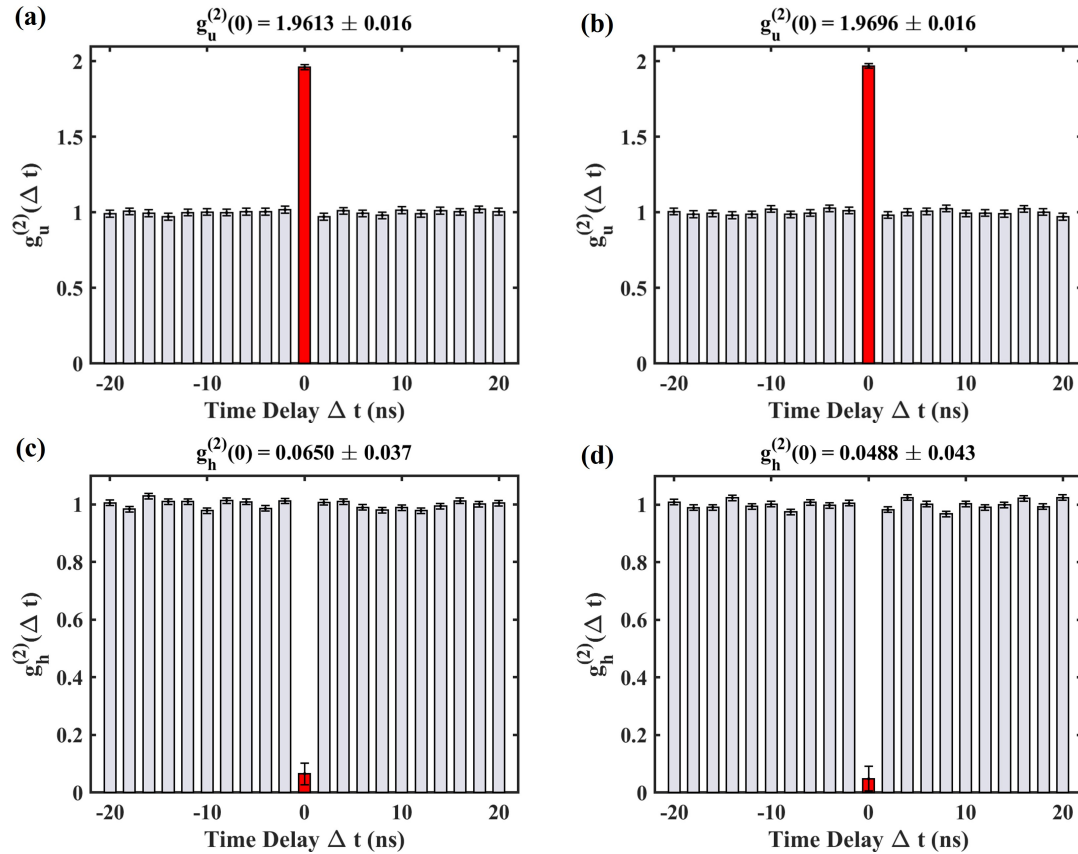


Fig. 2: Second-order correlation function measurements of the bidirectional integrated single-photon source. (a) and (b) Measured unheralded second-order correlation functions, $g_u^{(2)}(\Delta t)$, for the clockwise (CW) and counter-clockwise (CCW) directions, respectively. The high zero-delay peaks near 2 indicate the highly thermal nature of the individual fields and high spectral purities. (c) and (d) Measured heralded second-order correlation functions, $g_h^{(2)}(\Delta t)$, for the CW and CCW directions, respectively. The zero-delay dips significantly below 0.5 demonstrate the high-quality single-photon emission characteristics of both channels.

To evaluate the indistinguishability between the photons generated from the two spatial channels, we measured the two-photon interference visibility. Under the dual-pulse excited high-purity condition, the bidirectional single-photon source exhibits a remarkable two-photon interference visibility of 95.89%, as shown in Fig. 3. This high visibility confirms the exceptional consistency between the high-spectral-purity photon pairs generated from the two counter-propagating directions. For comparison, under standard single-pulse excitation, the visibility is measured to be 91.22%.

Conclusions

In summary, we have experimentally demonstrated a highly indistinguishable, dual-channel integrated single-photon source utilizing a bidirectional pumping architecture on a CMOS-compatible silicon platform. By exploiting the intrinsic symmetry of a single microring resonator, this approach inherently overcomes fabrication-induced resonance mismatches. Consequently, it significantly alleviates the need for complex active thermal tuning, significantly reducing system power consumption and thermal crosstalk. The device exhibits exceptional performance, achieving bidirectional spectral purities 96.13% and 96.96%,

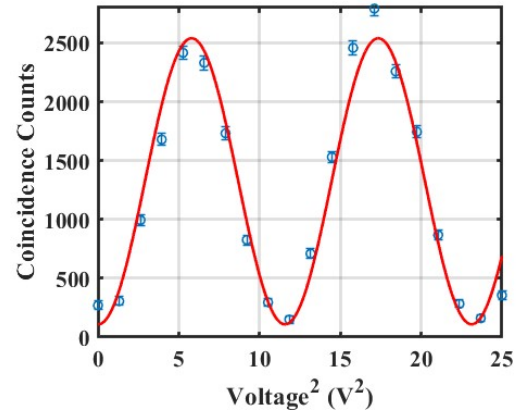


Fig. 3: On-chip two-photon interference visibility measurement under high-purity dual-pulse excitation. Error bars denote Poissonian counting uncertainties.

a low heralded $g_h^{(2)}(0)$ of 0.0488, and a remarkable two-photon interference visibility of 95.89%. Crucially, this robust architecture effectively mitigates the generation rate-loss penalties typical of standard pulse-shaping schemes. Looking forward, this scalable, tuning-free design paves the way for generating complex multi-photon entangled states.

Acknowledgements

This work was supported in part by the National Key Research and Development Program of China (2024YFE0211800), National Natural Science Foundation of China (U22A2082); Jinhua Science and Technology Program (2024-1-021); Shaoxing Industrial Technology Research and Development Projects (2024B1009); Ningbo Science and Technology Program (2023Z073 and 2024QL048); "Leading Goose" R&D Program of Zhejiang Province (2024C01112); "Vanguard" R&D Program of Zhejiang Province (2025C01043); "Vanguard" R&D Program of Zhejiang Province (2025C01009); the National Natural Science Foundation of China (No. 6250030072).the National Natural Science Foundation of China (No. 62505271); Zhejiang Provincial Natural Science Foundation of China (LMS26F050005); Leading Innovative and Entrepreneur Team Introduction Program of Zhejiang (2025R01004).

References

- [1] T. K. Paraiso, T. Roger, D. G. Marangon, I. De Marco, M. Sanzaro, R. I. Woodward, J. F. Dynes, Z. Yuan, A. J. Shields, "A photonic integrated quantum secure communication system", *Nature photonics*, vol. 15, no. 11, pp. 850–856, 2021. DOI: [10.1038/s41566-021-00873-0](https://doi.org/10.1038/s41566-021-00873-0).
- [2] S. Paesani, M. Borghi, S. Signorini, A. Mainos, L. Pavesi, A. Laing, "Near-ideal spontaneous photon sources in silicon quantum photonics", *Nature communications*, vol. 11, no. 1, p. 2505, 2020. DOI: [10.1038/s41467-020-16187-8](https://doi.org/10.1038/s41467-020-16187-8).
- [3] L. G. Helt, Z. Yang, M. Liscidini, J. E. Sipe, "Spontaneous four-wave mixing in microring resonators", *Optics letters*, vol. 35, no. 18, pp. 3006–3008, 2010. DOI: [10.1364/OL.35.003006](https://doi.org/10.1364/OL.35.003006).
- [4] I. I. Faruque, G. F. Sinclair, D. Bonneau, J. G. Rarity, and M. G. Thompson, "On-chip quantum interference with heralded photons from two independent micro-ring resonator sources in silicon photonics", *Optics express*, vol. 26, no. 16, pp. 20 379–20 395, 2018.
- [5] B. Hou, D. Chen, H. Ma, Z. Zhao, Q. Liang, B. Zhang, L. Ye, Q. Huang, D. Wang, T. Dai, H. Zhu, Y. Wang, J. Yang., "Chip-scale high spectral purity micro-ring photon-pair source based on dual-pulse control", *Optics Express*, vol. 34, no. 7, pp. 11 742–11 751, 2026. DOI: [10.1364/OE.589347](https://doi.org/10.1364/OE.589347).
- [6] S. Abel, F. Eltes, J. E. Ortmann, A. Messner, P. Castera, T. Wagner, D. Urbonas, A. Rosa, A. M. Gutierrez, D. Tulli, P. Ma, B. Baeuerle, A. Josten, W. Heni, D. Caimi, L. Czornomaz, A. A. Demkov, J. Leuthold, P. Sanchis, J. Fompeyrine, "Large pockels effect in micro-and nanostructured barium titanate integrated on silicon", *Nature materials*, vol. 18, no. 1, pp. 42–47, 2019. DOI: [10.1038/s41563-018-0208-0](https://doi.org/10.1038/s41563-018-0208-0).
- [7] J. B. Christensen, J. G. Koefoed, K. Rottwitt, and C. McKinstrie, "Engineering spectrally unentangled photon pairs from nonlinear microring resonators by pump manipulation", *Optics letters*, vol. 43, no. 4, pp. 859–862, 2018. DOI: [10.1364/OL.43.000859](https://doi.org/10.1364/OL.43.000859).
- [8] L. Rodda, B. M. Burrige, J. Barreto, I. I. Faruque, "Effect on the spectral purity of photon-pairs due to on-chip temporal manipulation of the pump", *APL Quantum*, vol. 1, no. 4, 2024. DOI: [10.1063/5.0229514](https://doi.org/10.1063/5.0229514).
- [9] B. Hou, H. Ma, Z. Zhao, Q. Liang, F. Ruan, Q. Huang, D. Chen, B. Zhang, T. Dai, Y. Wang, H. Zhu, J. Yang., "Silicon bidirectional-pumping-assisted multiplexing single-photon sources with high count rates", *Optics Letters*, vol. 51, no. 6, pp. 1661–1664, 2026. DOI: [10.1364/OL.586532](https://doi.org/10.1364/OL.586532).